

Assignment 4

(Questions from Module 6)

Name 1	
Matric No.	
Section	
Date	

Name 2	
Matric No.	
Section	
Date	

Total Marks: 20

Instructions

- Students must answer **ALL questions** and show complete workings clearly. Final answers without workings will not be accepted.
- Complete this assignment in **pairs (2 members ONLY)** and **submit handwritten** work in **PDF format** through eLearning within 7 days, as specified by your instructor.

Collaboration Requirements

Work together with your assigned partner and ensure both students contribute equally to the assignment. Communicate clearly and divide tasks fairly from the start.

Submission Format

Write neatly by hand in black or blue ink on standard A4 paper, then scan and convert to PDF. Include your names, student ID numbers, course details, instructor name, and submission date on the cover page.

Deadline and Submission

Confirm your specific 7-day deadline with your instructor. Submit only through the eLearning platform and keep a personal copy for your records.

QUESTION 1 (8 marks)

Main memory is divided into blocks. If the memory word is 32 bit and the size of a block is 4K words.

- i. Find memory capacity if total blocks = 1024 blocks.
- ii. Find number of blocks if main memory capacity is 8Mbit.

QUESTION 2 (12 marks)

A set associative cache size of 32K line divided into 2-line sets (2-way) with block a line of 4 words. Main memory capacity is 64M byte.

- i. How many bits are used for main memory address?
- ii. How many seats does the cache have?
- iii. Show the format of main memory addresses with tag, set and word bits.
- iv. What is main memory address (in hexadecimal) for the following cache data?

Tag	Set	word 11	word 10	word 01	word 00
2BAh	3B5Dh		Data		